

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.*
(BL2, BL3)

Activity Tool : Case Study (Medium)
Assessment Tool Weightage: 30%

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Analyze the case: An online shopping platform stores Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price). Identify FDs and normalize to 3NF.	BL2 – Understand	5
2	A hospital maintains Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName). Identify anomalies and normalize to BCNF.	BL3 – Apply	5
3	Case study: A university stores Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade). Analyze FDs and apply 2NF decomposition.	BL3 – Apply	5
4	Given the airline booking case: Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date). Normalize up to BCNF.	BL3 – Apply	5
5	Analyze an HR system schema for a multinational company. Identify all functional and transitive dependencies and normalize to 3NF.	BL2 – Understand	5
6	Case: Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate). Identify MVDs and decompose to 4NF.	BL3 – Apply	5

7	Analyze a telecom billing schema and identify all candidate keys, primary keys, and functional dependencies before normalization.	BL2 – Understand	5
8	A retail inventory system has Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse). Apply complete normalization.	BL3 – Apply	5
9	Study the given poorly normalized schema for a School ERP and propose a fully normalized design up to BCNF with justification.	BL3 – Apply	5
10	Given a movie streaming platform schema, identify join dependencies and decompose to 5NF with lossless-join verification.	BL3 – Apply	5

Code of Conduct:

I Y.Harsha192511368 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Y.Harsha

Q1. Online Shopping Platform – Normalize to 3NF

Given:

Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price)

Step 1: Functional Dependencies

Assume:

- $\text{OrderID} \rightarrow \text{CustID}$
- $\text{CustID} \rightarrow \text{CustName}$
- $\text{ProdID} \rightarrow \text{ProdName, Price}$
- $(\text{OrderID}, \text{ProdID}) \rightarrow \text{Qty}$

Therefore:

Candidate Key = (OrderID, ProdID)

Step 2: 1NF

All attributes contain atomic values, so the relation is in **1NF**.

Step 3: 2NF

There are partial dependencies:

- $\text{OrderID} \rightarrow \text{CustID}$
- $\text{ProdID} \rightarrow \text{ProdName, Price}$

Decompose into:

CUSTOMER

CustID CustName

CustID CustName

PRODUCT

ProdID ProdName Price

ProdID ProdName Price

ORDER

OrderID CustID

OrderID CustID

ORDER_DETAIL

OrderID ProdID Qty

OrderID ProdID Qty

Step 4: 3NF

Since:

$\text{CustID} \rightarrow \text{CustName}$

there is a transitive dependency in the original relation.

The above decomposition removes the transitive dependency.

Final 3NF Relations

- **CUSTOMER(CustID, CustName)**
- **PRODUCT(ProdID, ProdName, Price)**
- **ORDER(OrderID, CustID)**
- **ORDER_DETAIL(OrderID, ProdID, Qty)**

Conclusion: The decomposition removes redundancy and insertion, deletion and update anomalies.

Q2. Hospital – Normalize to BCNF

Given:

Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName)

Functional Dependencies

Assume:

- $PID \rightarrow PName, DocID, WardNo$
- $DocID \rightarrow DocName, Specialization$
- $WardNo \rightarrow WardName$

Anomalies

1. Update anomaly:

If a doctor's specialization changes, multiple patient records may need updating.

2. Insertion anomaly:

A new doctor cannot be stored unless a patient is assigned.

3. Deletion anomaly:

Deleting the last patient of a doctor may remove information about that doctor.

BCNF Decomposition

PATIENT

Patient(PID, PName, DocID, WardNo)

DOCTOR

Doctor(DocID, DocName, Specialization)

WARD

Ward(WardNo, WardName)

BCNF Check

- $PID \rightarrow PName, DocID, WardNo$; PID is a key.
- $DocID \rightarrow DocName, Specialization$; DocID is a key in DOCTOR.
- $WardNo \rightarrow WardName$; WardNo is a key in WARD.

Thus every determinant is a candidate key in its relation.

Conclusion: The decomposition satisfies **BCNF** and reduces redundancy.

Q3. University Enrollment – Apply 2NF

Given:

Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade)

Functional Dependencies

- $SID \rightarrow SName$
- $CourseID \rightarrow CourseName, Faculty$
- $(SID, CourseID) \rightarrow Grade$

Candidate Key

(SID, CourseID)

The combination uniquely identifies an enrollment record.

1NF

All attributes are atomic, so the table is in **1NF**.

Partial Dependencies

Because the key is composite:

- $SID \rightarrow SName$

- $\text{CourseID} \rightarrow \text{CourseName, Faculty}$

These are partial dependencies, so the relation is **not in 2NF**.

2NF Decomposition

STUDENT

Student(SID, SName)

COURSE

Course(CourseID, CourseName, Faculty)

ENROLLMENT

Enrollment(SID, CourseID, Grade)

Result

- Student details are stored once.
- Course details are stored once.
- Grade depends on both SID and CourseID.

Conclusion: The decomposition removes partial dependencies and achieves **2NF**.

Q4. Airline Booking – Normalize up to BCNF

Given:

Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date)

Functional Dependencies

Assume:

- $\text{PassID} \rightarrow \text{PassName}$
- $\text{FlightNo} \rightarrow \text{DeptCity, ArrCity}$
- $(\text{FlightNo, Date, PassID}) \rightarrow \text{Seat}$

Candidate key:

(FlightNo, Date, PassID)

1NF

All attributes contain atomic values $\rightarrow$ **1NF**.

2NF

Partial dependencies exist:

- $\text{PassID} \rightarrow \text{PassName}$
- $\text{FlightNo} \rightarrow \text{DeptCity, ArrCity}$

Decompose:

PASSENGER

Passenger(PassID, PassName)

FLIGHT

Flight(FlightNo, DeptCity, ArrCity)

BOOKING

Booking(FlightNo, Date, PassID, Seat)

BCNF

In each relation:

- PassID is a key of PASSENGER.
- FlightNo is a key of FLIGHT.
- (FlightNo, Date, PassID) is a key of BOOKING.

Therefore, each determinant is a candidate key.

Final BCNF Design

- **PASSENGER(PassID, PassName)**
- **FLIGHT(FlightNo, DeptCity, ArrCity)**
- **BOOKING(FlightNo, Date, PassID, Seat)**

Conclusion: The schema is normalized up to **BCNF**.

Q5. HR System – Normalize to 3NF

The question asks to analyze an HR schema and identify functional and transitive dependencies, but the uploaded document does **not provide the actual HR attributes/schema**.

A standard HR example can be represented as:

Employee(EmpID, EmpName, DeptID, DeptName, ManagerID, ManagerName)

Functional Dependencies

- $\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}$
- $\text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$
- $\text{ManagerID} \rightarrow \text{ManagerName}$

Transitive Dependencies

$\text{EmpID} \rightarrow \text{DeptID} \rightarrow \text{DeptName}$

Therefore:

$\text{EmpID} \rightarrow \text{DeptName}$

This is a transitive dependency.

Also:

$\text{EmpID} \rightarrow \text{DeptID} \rightarrow \text{ManagerID} \rightarrow \text{ManagerName}$

3NF Decomposition

EMPLOYEE

Employee(EmpID, EmpName, DeptID)

DEPARTMENT

Department(DeptID, DeptName, ManagerID)

MANAGER

Manager(ManagerID, ManagerName)

Result

The decomposition removes transitive dependencies and reduces redundancy.

Conclusion: The HR database is normalized to **3NF**.

This is a standard model because the assessment question does not give a specific HR schema.

Q6. Library – MVDs and 4NF

Given:

Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate)

Assume a member can independently borrow multiple books and have multiple genres associated with their library activity.

Functional Dependencies

- $\text{MemberID} \rightarrow \text{MemberName}$
- $\text{BookID} \rightarrow \text{Title}, \text{Author}, \text{Genre}$

- $(\text{MemberID}, \text{BookID}) \rightarrow \text{BorrowDate}$

Multivalued Dependency

If independent multiple values exist for a member:

MemberID $\twoheadrightarrow$ **BookID**

and

MemberID $\twoheadrightarrow$ **Genre**

This creates unnecessary combinations.

Example

A member has:

- Books: B1, B2
- Genres: Fiction, Science

The database may unnecessarily produce:

MemberID BookID Genre

M1 B1 Fiction

M1 B1 Science

M1 B2 Fiction

M1 B2 Science

4NF Decomposition

MEMBER

Member(MemberID, MemberName)

MEMBER_BOOK

MemberBook(MemberID, BookID, BorrowDate)

BOOK

Book(BookID, Title, Author, Genre)

Result

The independent multivalued facts are separated.

Conclusion: The decomposition eliminates MVD-based redundancy and satisfies **4NF**, under the stated independence assumption.

Q7. Telecom Billing – Keys and Functional Dependencies

The question asks to identify candidate keys, primary keys and FDs, but **the uploaded assessment does not provide the actual telecom billing schema.**

A standard telecom billing schema can be considered:

Billing(CustomerID, CustomerName, PhoneNo, PlanID, PlanName, CallID, CallDuration, CallCharge)

Functional Dependencies

- $\text{CustomerID} \rightarrow \text{CustomerName}, \text{PhoneNo}, \text{PlanID}$
- $\text{PlanID} \rightarrow \text{PlanName}$
- $\text{CallID} \rightarrow \text{CustomerID}, \text{CallDuration}, \text{CallCharge}$

Candidate Keys

Possible candidate keys:

- CustomerID
- CallID

depending on the business rules.

Primary Key

For the billing transaction relation:

Primary Key = CallID

Foreign Keys

- CustomerID references CUSTOMER.
- PlanID references PLAN.

Important Dependencies

CustomerID $\rightarrow$ CustomerName

CustomerID $\rightarrow$ PhoneNo

CustomerID $\rightarrow$ PlanID

PlanID $\rightarrow$ PlanName

CallID $\rightarrow$ CustomerID, CallDuration, CallCharge

Conclusion

Identifying keys and FDs before normalization helps detect **partial and transitive dependencies** and provides the basis for normalization.

Since the assessment gives no telecom schema, the above is a representative answer rather than a reconstruction of a missing schema.

Q8. Retail Inventory – Complete Normalization

Given:

Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse)

Functional Dependencies

Assume:

- PID $\rightarrow$ PName, SupplierID, Category, Price, Warehouse
- SupplierID $\rightarrow$ SupplierName

1NF

All values are atomic.

Therefore, the relation is in **1NF**.

2NF

If PID is the single candidate key, there are no partial dependencies.

Therefore, it satisfies **2NF**.

3NF

There is a transitive dependency:

PID $\rightarrow$ SupplierID $\rightarrow$ SupplierName

So SupplierName should not be stored in PRODUCT.

Decomposition

PRODUCT

Product(PID, PName, SupplierID, Category, Price, Warehouse)

SUPPLIER

Supplier(SupplierID, SupplierName)

BCNF

- PID is a key of PRODUCT.
- SupplierID is a key of SUPPLIER.

Therefore, determinants are candidate keys.

Final Normalized Design

PRODUCT(PID, PName, SupplierID, Category, Price, Warehouse)

SUPPLIER(SupplierID, SupplierName)

Conclusion: The schema is normalized through $1NF \rightarrow 2NF \rightarrow 3NF \rightarrow BCNF$.

Q9. School ERP – Normalize up to BCNF

The question asks for a poorly normalized School ERP design, but **the actual schema is not supplied in the assessment document**.

A standard School ERP relation can be considered:

School(SID, SName, DeptID, DeptName, CourseID, CourseName, FacultyID, FacultyName, Grade)

Functional Dependencies

- $SID \rightarrow SName, DeptID$
- $DeptID \rightarrow DeptName$
- $CourseID \rightarrow CourseName, FacultyID$
- $FacultyID \rightarrow FacultyName$
- $(SID, CourseID) \rightarrow Grade$

Problems

There are transitive dependencies:

$SID \rightarrow DeptID \rightarrow DeptName$

and

$CourseID \rightarrow FacultyID \rightarrow FacultyName$

There are also partial dependencies because the enrollment key is:

(SID, CourseID)

BCNF Decomposition

STUDENT

Student(SID, SName, DeptID)

DEPARTMENT

Department(DeptID, DeptName)

COURSE

Course(CourseID, CourseName, FacultyID)

FACULTY

Faculty(FacultyID, FacultyName)

ENROLLMENT

Enrollment(SID, CourseID, Grade)

BCNF Verification

- SID is a key of STUDENT.
- DeptID is a key of DEPARTMENT.
- CourseID is a key of COURSE.
- FacultyID is a key of FACULTY.
- (SID, CourseID) is a key of ENROLLMENT.

Therefore, all determinants are candidate keys.

Conclusion: The proposed design eliminates redundancy and satisfies **BCNF**.

This is a representative School ERP design because the original question does not provide the poorly normalized schema.

Q10. Movie Streaming Platform – 5NF

The question asks to identify join dependencies and decompose a movie-streaming schema to **5NF with lossless-join verification**, but the assessment document does **not provide the actual relation schema**.

A standard example can be:

Streaming(Movie, Actor, Platform)

Suppose a movie can have multiple actors and can be available on multiple platforms, and these relationships are independent.

Join Dependency

The relation may have a join dependency:

(Movie, Actor), (Movie, Platform), (Actor, Platform)

This means the complete relation can be reconstructed from smaller relations.

Decomposition

MOVIE_ACTOR

MovieActor(Movie, Actor)

MOVIE_PLATFORM

MoviePlatform(Movie, Platform)

ACTOR_PLATFORM

ActorPlatform(Actor, Platform)

5NF

A relation is in **5NF** when every non-trivial join dependency is implied by candidate keys.

The decomposition removes redundancy caused by join dependencies.

Lossless Join Verification

The original relation can be reconstructed using natural joins:

MovieActor $\bowtie$ MoviePlatform $\bowtie$ ActorPlatform

If the resulting relation contains exactly the original valid tuples without spurious tuples, the decomposition is **lossless**.

Conclusion

The decomposition into:

- MovieActor(Movie, Actor)
- MoviePlatform(Movie, Platform)
- ActorPlatform(Actor, Platform)

removes join dependency redundancy and provides a **5NF design**, assuming the stated independent relationships.